Q1. Create a New Database and Table for Employees Task: Create a new database namedcompany_db and Create a table named employee with the following columns:

Column Name	Data Type	Constraint
employee_id	INT	PRIMARY KEY
first_name	VARCHAR(50)	
last_name	VARCHAR(50)	
department	VARCHAR(50)	
salary	INT	
hire_date	DATE	

create database company_db;

use company_db;

create table employees(

employee_id int primary key,

first_name varchar(50) NOT NULL,

last_name varchar(50) NOT NULL,

department varchar(50) NOT NULL,

salary int,

hire_date date);

Q2. Insert Data into Employees Table Task: Insert the following sample records into the employess table

employee_id	first_name	last_name	department	salary	hire_date
101	Amit	Sharma	HR	50000	2020-01-15
102	Riya	Kapoor	Sales	75000	2019-03-22
103	Raj	Mehta	IT	90000	2018-07-11
104	Neha	Verma	IT	85000	2021-09-01
105	Arjun	Singh	Finance	60000	2022-02-10

insert into employees(employee_id,first_name,last_name,department,salary,hire_date)

values

(101,'Amit','Sharma','HR',50000,'2020-01-15'),

(102,'Riya','Kapoor','Sales',75000,'2019-03-22'),

(103,'Raj','Mehata','IT',90000,'2018-07-11'),

(104,'Neha','Verma','IT',85000,'2021-09-01'),

(105,'Arjun','Singh','Finance',60000,'2022-02-10');

Q3. Display All Employee Records Sorted by Salary (Lowest to Highest)

Select*from employees

Order by salary asc;

Q4. Show Employees Sorted by Department (A–Z) and Salary (High → Low)

select* from employees

order by department asc , salary desc;

Q5. List All Employees in the IT Department, Ordered by Hire Date (Newest First)

select first_name,last_name,hire_date

from   employees

where department = 'IT'

ORDER BY hire_date asc;

Q6. Create and Populate a Sales Table Task: Create a table sales to track sales data:

sale_id	customer_name	amount	sale_date
1	Aditi	1500	2024-08-01
2	Rohan	2200	2024-08-03
3	Aditi	3500	2024-09-05
4	Meena	2700	2024-09-15
5	Rohan	4500	2024-09-25

create table sales(

sale_id int primary key,

```
customer_name varchar(50),
amount varchar(50),
sale_date date);
insert into sales(sale_id,customer_name,amount,sale_date)
values
(1,'Aditi',1500,'2024-08-01'),
(2,'Rohan',2200,'2024-08-03'),
(3,'Aditi',3500,'2024-09-05'),
(4,'Meena',2700,'2024-09-15'),
(5,'Rohan',4500,'2024-09-25');
```

Q7. Display All Sales Records Sorted by Amount (Highest → Lowest)

```
select * from sales
order by amount desc;
```

Q8. Show All Sales Made by Customer "Aditi

```
select *from sales
where customer_name='Aditi';
```

Q9. What is the Difference Between a Primary Key and a Foreign Key?

Aspect	Primary Key	Foreign Key
Purpose	Uniquely identifies each record in a table	Establishes a relationship between two tables
Uniqueness	Must be unique	Can have duplicate values
NULL values	Cannot contain NULL values	Can contain NULL values (unless restricted)
Number per table	Only one primary key per table	A table can have multiple foreign keys
Location	Defined in the same table it identifies	Defined in one table but references a primary key in another table

Data integrity	Ensures entity integrity	Ensures referential integrity
Example	StudentID in Students table	StudentID in Enrollments table referencing Students(StudentID)

Q10. What Are Constraints in SQL and Why Are They Used?

Constraints in SQL are rules applied to database tables or columns that restrict the data that can be inserted, updated, or deleted. They ensure the accuracy, consistency, and reliability of data in a database.

Why Are Constraints Used?

- To maintain data integrity
- To prevent invalid or inconsistent data
- To enforce business rules
- To ensure proper relationships between tables

Constraint	Purpose
NOT NULL	Prevents NULL values
UNIQUE	Ensures all values are different
PRIMARY KEY	Uniquely identifies each record
FOREIGN KEY	Links tables and maintains referential integrity
CHECK	Validates data based on a condition
DEFAULT	Sets a default value